



Universidad
Rey Juan Carlos

HIGHER TECHNICAL SCHOOL OF COMPUTER ENGINEERING
MASTER'S DEGREE IN COMPUTER GRAPHICS, GAMES, AND VIRTUAL
REALITY

2025-2026 Academic Year

Advanced Rendering I

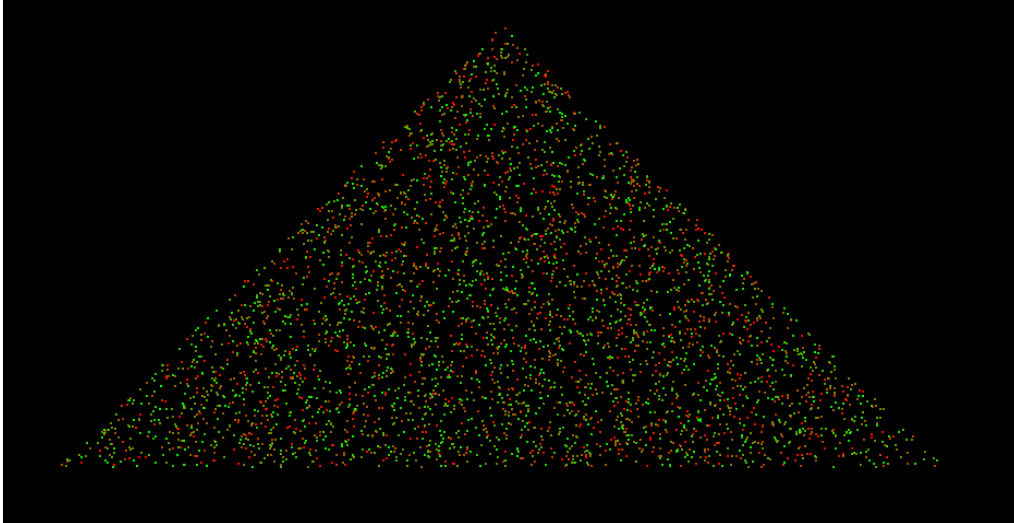
Practice 1

Carlos Escribano González

Sampling:

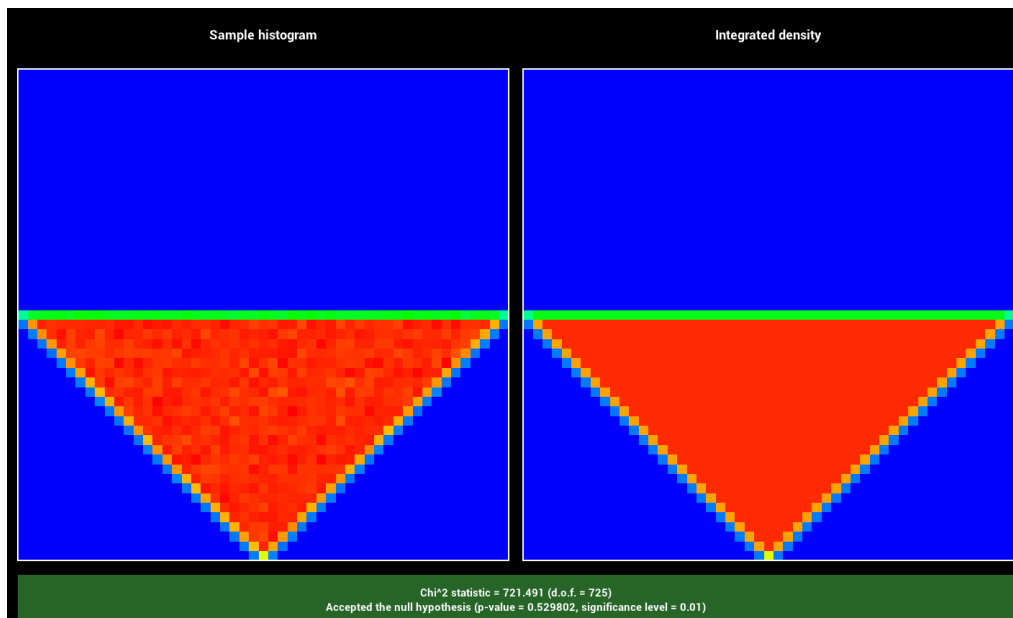
Tent Triangle Sampling:

After warping the input sample using the *Warp::squareToTent()* function to ensure that the sum of its coordinates is less than or equal to 1, its values were used as barycentric coordinates to generate samples within the isosceles triangle described in the problem statement. The resulting distribution is shown below:



Triangle sampling

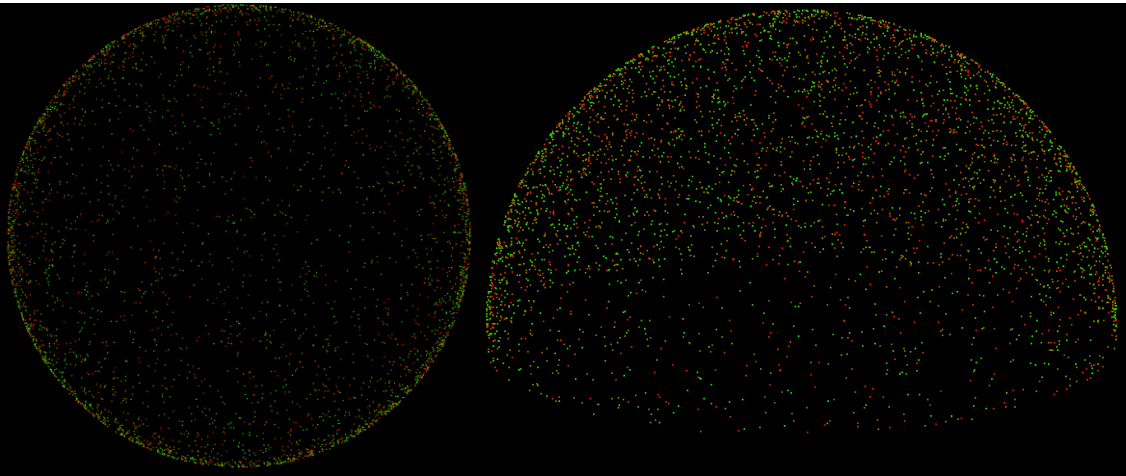
The resulting distribution is consistent with uniform sampling over the triangle, as confirmed by the null hypothesis test:



Null hypothesis test for triangle sampling

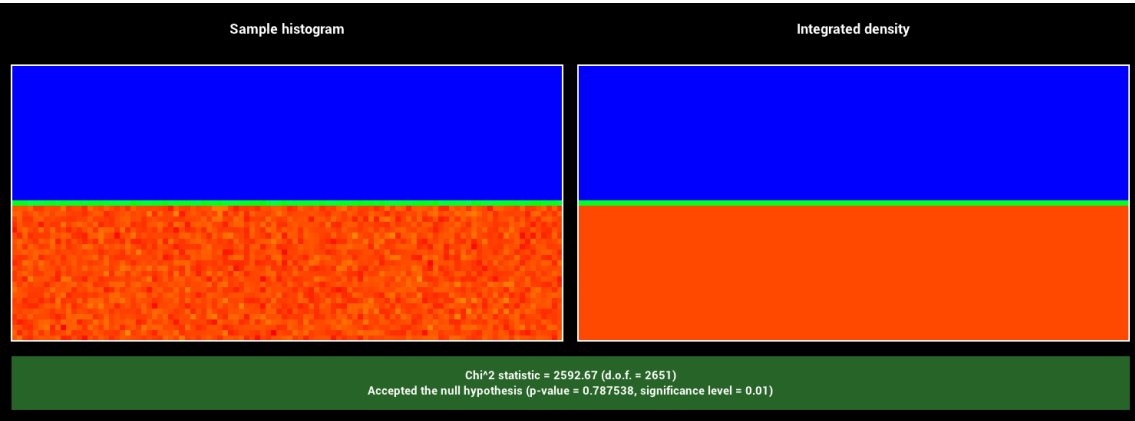
Uniform Hemisphere Sampling:

After converting the input sample values into spherical coordinates, uniformly distributed vectors over the hemisphere are obtained, as shown below:



Uniform hemisphere sampling

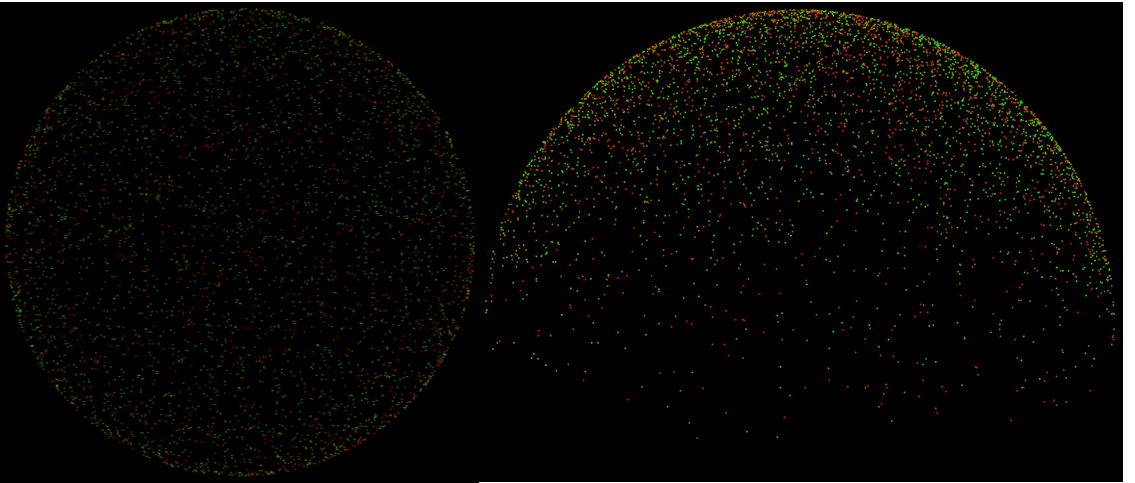
The result was confirmed using the null hypothesis test:



Null hypothesis test for uniform hemisphere sampling

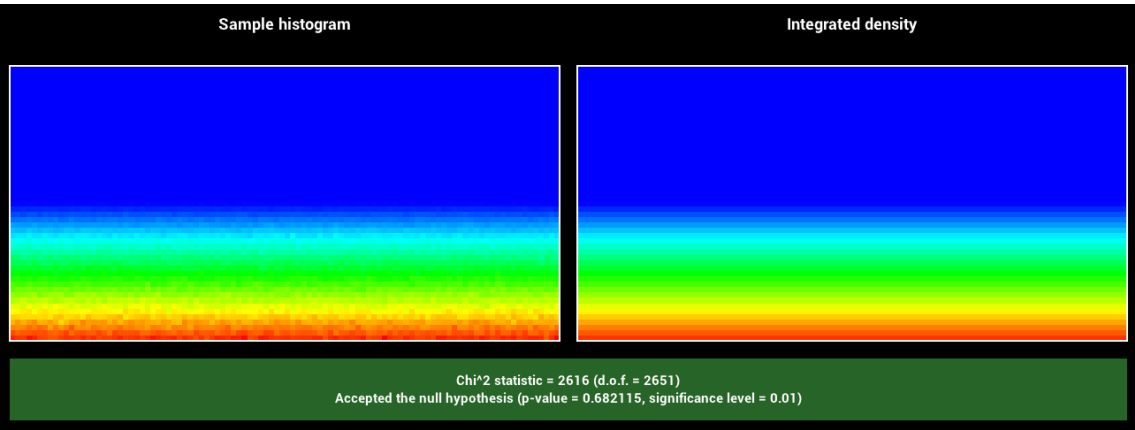
Cosine-Weighted Hemisphere Sampling:

Before performing cosine-weighted hemisphere sampling, a uniform disk was warped using concentric mapping. The resulting disk samples were then projected onto the hemisphere according to Malley's method to obtain cosine-weighted hemisphere samples:



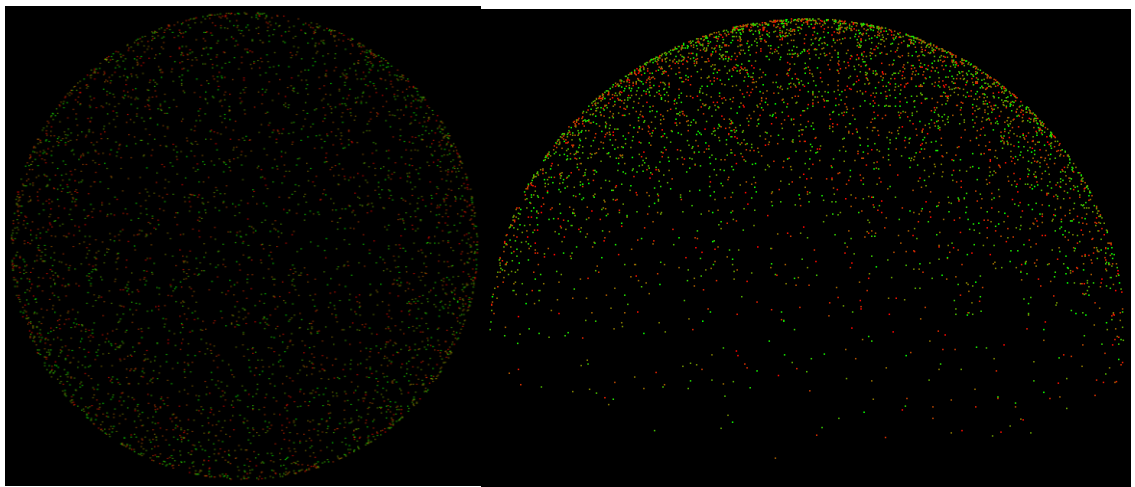
Cosine-weighted hemisphere sampling (Malley)

Once again, the resulting distribution was analyzed using the null hypothesis test, which was satisfied.



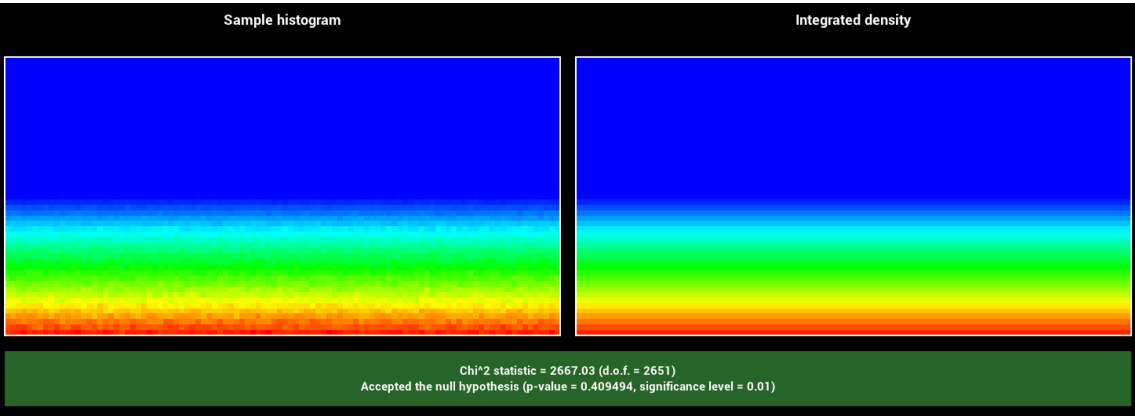
Null hypothesis test for cosine-weighted hemisphere sampling (Malley)

Additionally, inverse transform sampling was performed, following a procedure similar to that used for uniform hemisphere sampling, but using a different distribution for theta. In this case, theta is sampled according to the inverse CDF of the corresponding cosine-weighted distribution:



Cosine-weighted hemisphere sampling (inverse transform sampling)

The resulting distribution is consistent with that obtained using Malley's method and was also validated by the null hypothesis test:



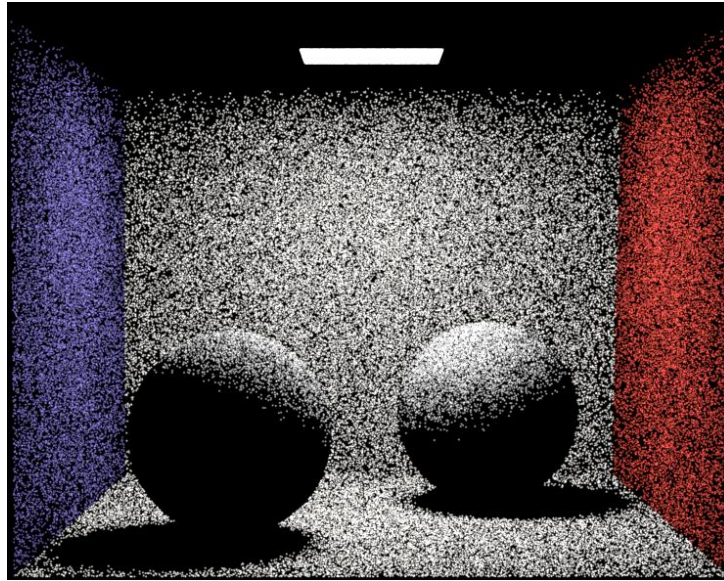
Null hypothesis test for cosine-weighted hemisphere sampling (inverse transform sampling)

Both methods are equivalent and produce the same distribution.

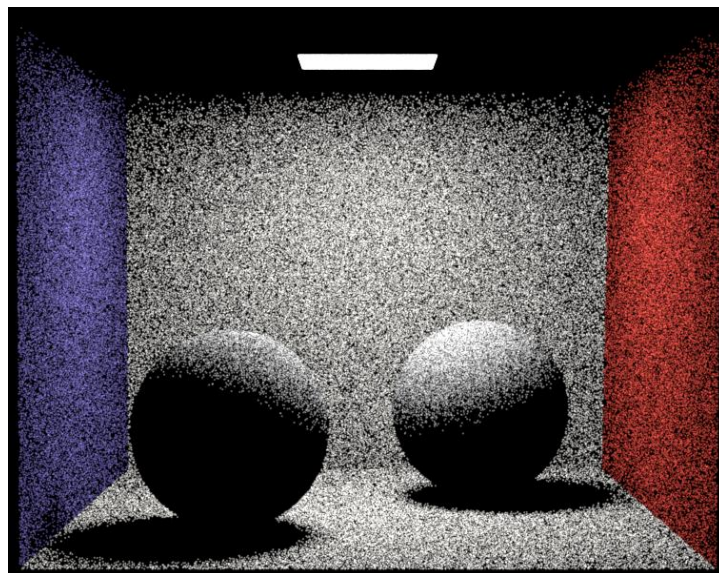
Direct Lighting with Material Sampling:

In the second part of the practice, an integrator based on sampling Lambertian materials under direct lighting was implemented. Each ray cast from the camera that intersects an object in the scene checks whether the intersected object is an emitter. If it is, the emitted radiance is returned directly. Otherwise, the material's BSDF is sampled, and a secondary ray is traced. If the mesh intersected by the secondary ray is an emitter, its radiance is used to multiply the value previously calculated from the sample; otherwise, black is returned.

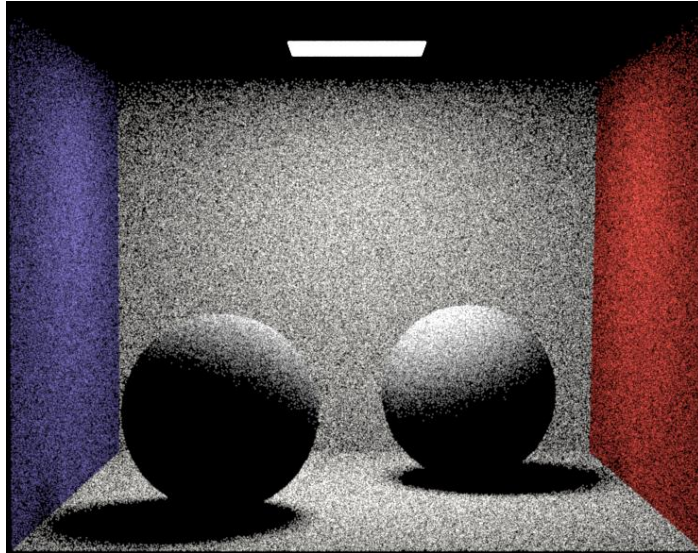
The results obtained for the *cbox* scene are shown below:



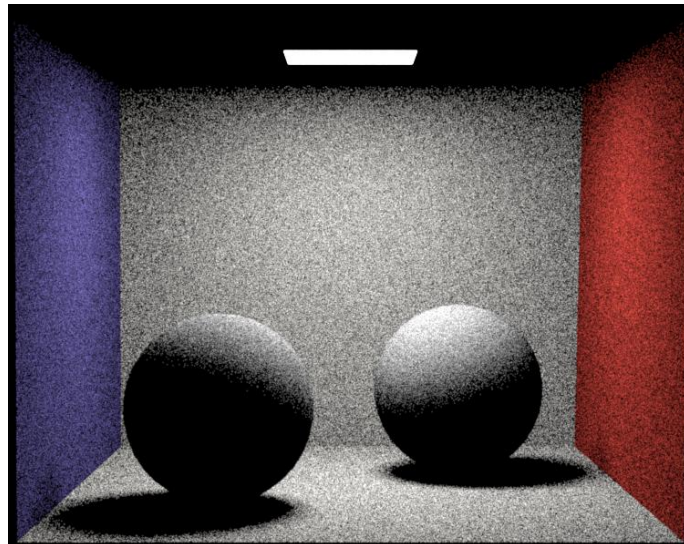
"cbox" scene with 32 samples per pixel



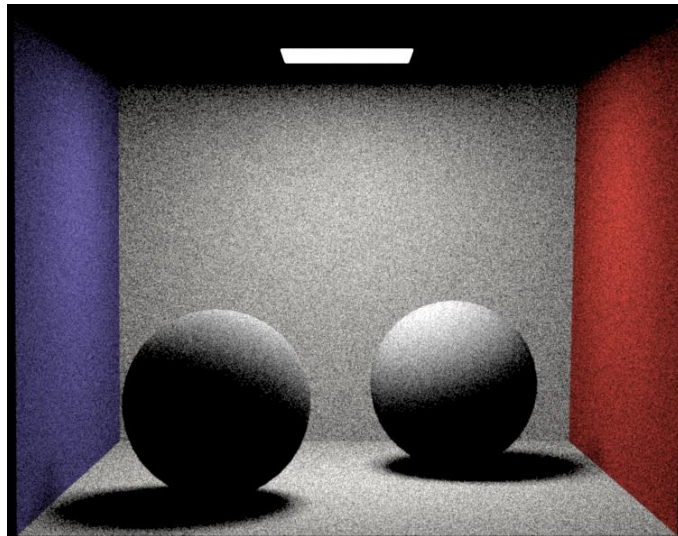
"cbox" scene with 64 samples per pixel



"cbox" scene with 128 samples per pixel



"cbox" scene with 256 samples per pixel



"cbox" scene with 512 samples per pixel

It can be observed that as the number of samples per pixel increases, the reduction in noise becomes less noticeable. This is consistent with Monte Carlo theory, since the error decreases at a rate of $\frac{1}{\sqrt{N}}$, meaning that reducing the error by half requires quadrupling the number of samples.

For the *motto* scene, which contains two lights with different colors, some of the results obtained are shown below:



"motto" scene with 32 samples per pixel



"motto" scene with 128 samples per pixel



"motto" scene with 512 samples per pixel